

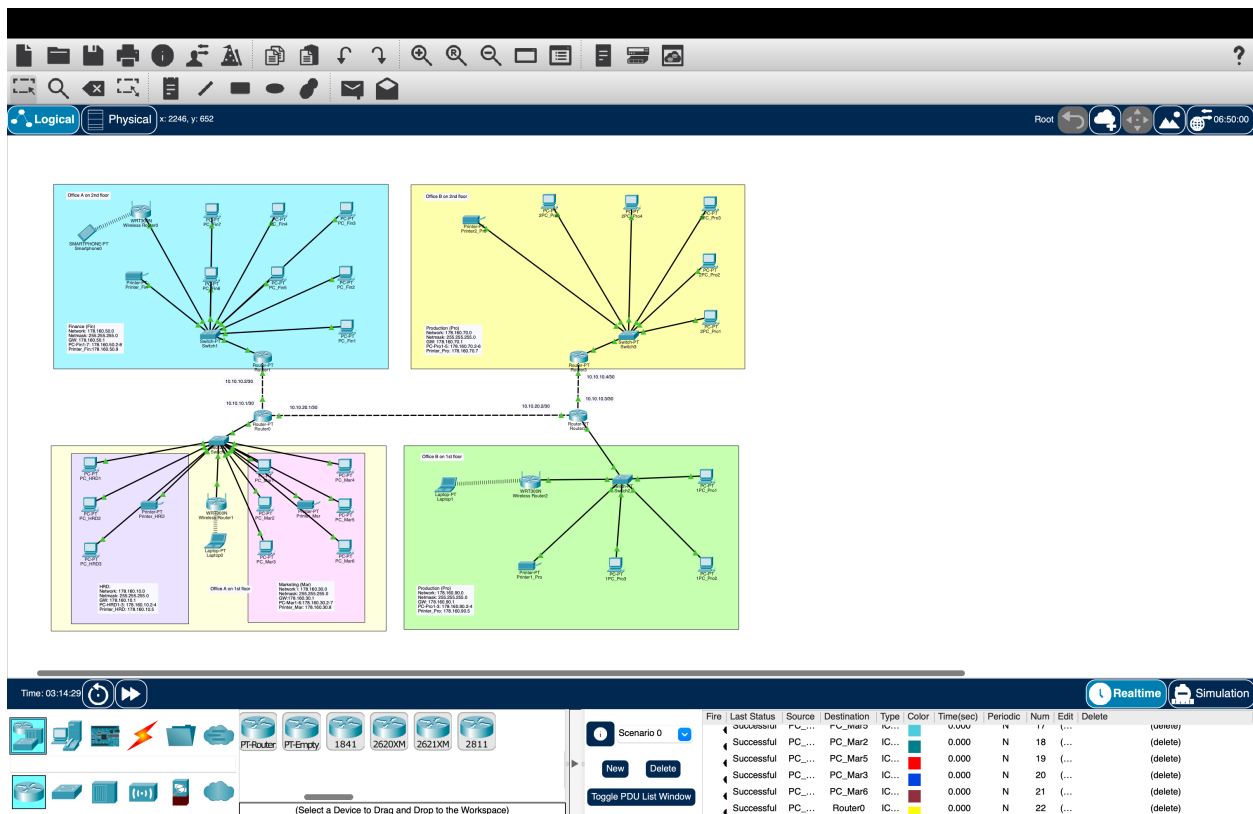
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CLASS : IT4

COMPUTER NETWORK

- Design Topology



- Switch A Configuration

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HRD
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name Marketing
Switch(config-vlan)#int fa0/12
Switch(config-if)#switchport mode trunk
Switch(config-if)#exit
Switch(config)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config)#int range fa0/1 - 4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#
%LINK-3-UPDOWN: Interface FastEthernet0/11, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up

Switch(config)#int range fa0/5 - 11
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#int
%LINK-3-UPDOWN: Interface FastEthernet0/12, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed state to up

% Incomplete command.
Switch(config)#int fa0/12
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#

```

- Router 0 Configuration

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 178.160.10.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#int f0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 178.160.30.1 255.255.255.0
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#int f0/0
Router(config-if)#exit
Router(config)#int f0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.30, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 178.160.10.1 178.160.10.2
      ^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 178.160.10.1 178.160.10.2
Router(config)#ip dhcp pool HRD
Router(dhcp-config)#default-router 178.160.10.1
Router(dhcp-config)#network 178.160.10.0 255.255.255.0
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 178.160.30.1
Router(config)#ip dhcp pool Marketing
Router(dhcp-config)#default-router 178.160.30.1
Router(dhcp-config)#network 178.160.30.0 255.255.255.0
Router(dhcp-config)#exit
Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router(config-if)#int f2/0
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up

Router(config-if)#ip route 178.160.50.0 255.255.255.0 10.10.10.2
Router(config)#ip route 178.160.70.0 255.255.255.0 10.10.20.2
Router(config)#ip route 178.160.90.0 255.255.255.0 10.10.20.2
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

```

- Router 1 Configuration

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 178.160.50.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 178.160.50.1
Router(config)#ip dhcp pool Finance
Router(dhcp-config)#default-router 178.160.50.1
Router(dhcp-config)#network 178.160.50.0 255.255.255.0
Router(dhcp-config)#exit
Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.2255.255.255.0
                        ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 10.10.10.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config-if)#ip route 0.0.0.0 0.0.0.0 10.10.10.1
Router(config)#ip route 178.160.10.0 255.255.255.0 10.10.10.1
Router(config)#ip route 178.160.30. 255.255.255.0 10.10.10.1
                        ^
% Invalid input detected at '^' marker.

Router(config)#ip address 178.160.30.0 255.255.255.0 10.10.10.1
                        ^
% Invalid input detected at '^' marker.

Router(config)#ip address 178.160.30.0 255.255.255.0 10.10.10.1
                        ^
% Invalid input detected at '^' marker.

Router(config)#ip route 178.160.30.0 255.255.255.0 10.10.10.1
Router(config)#ip route 178.160.70.0 255.255.255.0 10.10.10.1
Router(config)#ip route 178.160.90.0 255.255.255.0 10.10.10.1
Router(config)#
```

- Router 2 Configuration

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 178.160.90.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 178.160.90.1
Router(config)#ip dhcp pool Production
Router(dhcp-config)#default-router 178.160.90.1
Router(dhcp-config)#network 178.160.90.0 255.255.255.0
Router(dhcp-config)#exit
Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.4 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config-if)#ip route 0.0.0.0 0.0.0.0 10.10.10.3
Router(config)#ip route 178.160.70 255.255.255.0 10.10.10.3
                        ^
% Invalid input detected at '^' marker.

Router(config)#ip route 178.160.70.0 255.255.255.0 10.10.10.3
Router(config)#ip route 178.160.10.0 255.255.255.0 10.10.10.3
Router(config)#ip route 178.160.10.0 255.255.255.0 10.10.10.3
Router(config)#ip route 178.160.50.0 255.255.255.0 10.10.10.3
Router(config)#

Router>end
Translating "end"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.4 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#int f2/0
Router(config-if)#ip address 10.10.20.0 255.255. 255.0
                        ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 10.10.20.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up

```

- Router 3 Configuration

```

Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int f0/0
Router(config-if)#ip address 178.160.70.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 178.160.70.1
Router(config)#ip dhcp pool Production
Router(dhcp-config)#default-router 178.160.70.1
Router(dhcp-config)#network 178.160.70.0 255.255.255.0
Router(dhcp-config)#exit
Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.3 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

Router(config-if)#int f2/0
%Invalid interface type and number
Router(config)#ip address 10.10.20.2 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config)#ip address 10.10.20.2 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config)#ip address 10.10.10.3 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config)#ip address 10.10.20.2 255.255.255.0
      ^
% Invalid input detected at '^' marker.

Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.3 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#int f2/0
%Invalid interface type and number
Router(config)#int f1/0
Router(config-if)#ip address 10.10.10.3 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#int f2/0
%Invalid interface type and number
Router(config)#int fa2/0
%Invalid interface type and number
Router(config)#

```